DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

948 ways to root

Issue #38 | Fri, 06 Nov 2026 | ☕ ~9 min read

⚡ If you only have one minute

1. **OpenAI's agents found 948 Linux kernel bugs** that turn a normal user into root, at about **$1,280 each**. Patches are rolling out; keep your servers updated.
2. **ChatGPT now makes animated GIFs.** Images 2.6 is rolling out on every ChatGPT and Codex plan, and OpenAI says it is 50% faster.
3. **Part 1: scaling laws and benchmarks.** How to read the charts in every model launch without getting fooled.

Today in three numbers

948
distinct Linux **root bugs** OpenAI's agents exploited in one October run

4B
images a week made with ChatGPT, OpenAI **claims**, up from 3B+ in September

4
rides booked by **AI agents** in Uber's busiest October week, out of ~320M trips and orders

PART 1 Learn one thing

📈 Scaling laws and benchmarks: how to read model launch charts

4 MIN READ | MODULE 5 · UNDER THE HOOD

Every model launch comes with a chart where the new model's bar is tallest. Scaling laws tell you why models get better; a few simple questions tell you whether the chart means anything.

The plain-English version. A **scaling law** is a pattern researchers found: if you give a model **more parameters, more training data and more compute**, its error falls in a smooth, predictable curve. That predictability is why labs dare to spend billions on one training run. They train small models first, draw the curve, and **extrapolate** to the big one.

The famous lesson is Chinchilla (DeepMind, 2022). DeepMind's 280B-parameter **Gopher** had been trained on 300B tokens. A **70B model, Chinchilla, trained on 1.4 trillion tokens** with about the same compute, beat it on many benchmarks while being **4 times smaller**. (Alexandra Barr) The takeaway: **grow data and model size together**, roughly **20 training tokens per parameter**. (Knowledge Space)

🐘 Gopher (2021)

- **280B** parameters
- **300B** training tokens
- About 1 token per parameter: **undertrained**

```
big brain, short school
```

VS

🐭 Chinchilla (2022)

- **70B** parameters
- **1.4T** training tokens
- 20 tokens per parameter: **compute-optimal**

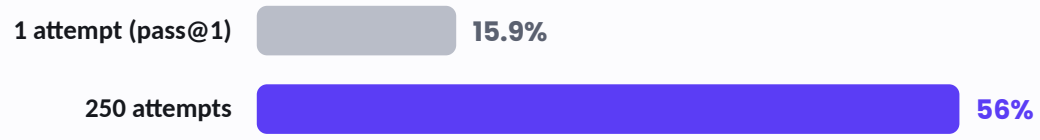
```
smaller brain, longer school
```

Same training budget (about 10^24 FLOPs, per Barr's explainer). Chinchilla won on many benchmarks and is also cheaper to run, because inference cost follows size.

Analogy: two students with the same tuition budget. One spends it on a huge library and barely reads; the other buys fewer books and **reads every one of them five times**. The second student does better in the exam. Today's labs go even further: they **"over-train" small models** on far more than 20 tokens per parameter, because a small, well-read model is cheap to serve to millions of users.

Now the exam itself: benchmarks. A benchmark is a fixed test set (maths problems, GitHub issues, terminal tasks) with a score. The trouble is that **the same model can get very different scores** depending on how it is tested. One review found SWE-bench Lite going from **15.9% with one attempt to 56% with 250 attempts**, and a **76-point swing** on one task just from changing the prompt format on LLaMA-2-13B. (IntuitionLabs)

Same benchmark, different rules: SWE-bench Lite



Figures as reported by IntuitionLabs' benchmark guide (Sep 2026). Same task set, very different headline number.

So before you trust a launch chart, ask five questions:

1. **Who ran it?** The company's own numbers are **claims** until someone else reproduces them.
2. **How many tries?** pass@1 (one shot) and "best of many" are not the same test.
3. **What harness and tools?** Agent scores depend heavily on the scaffolding around the model (Issue #26).
4. **Is the test fresh?** If the questions leaked into training data (**contamination**), the score measures memory, not skill.
5. **Is the gap bigger than the noise?** A 1-point lead on a few hundred questions is often a tie.

Why now? This week alone, Ricursive's open **Ripple-65B** claims it beats much larger open models at coding, and Kimi K3.1 Pro shipped with **only Moonshot's own scores** (Issue #37). (The Neuron) Those five questions are how you judge both.

📖 Jargon buster

- **Compute (FLOPs):** the total number of maths operations used to train. A handy rule: **about 6 × parameters × tokens**.
- **Compute-optimal:** the best model you can get **for a fixed training budget**.
- **pass@k:** the chance that **at least one of k attempts** passes. pass@1 is the honest default.
- **Contamination:** benchmark questions **leaking into the training data**, inflating the score.

🧪 Try it in 10 minutes

- **No code:** open a recent launch post and check the benchmark small print for **attempts, tools and who ran it**. Then find an independent result.
- **Some code:** run the snippet below: see how fast pass@k climbs for a model that solves 2 in 10.

🛠️ For devs: compute pass@k properly

```
from math import comb
def pass_at_k(n, c, k):   # n samples, c correct
    if n - c < k: return 1.0
    return 1 - comb(n - c, k) / comb(n, k)
for k in (1, 5, 10):
    print(k, round(pass_at_k(10, 2, k), 3))
# 1 0.2
# 5 0.778
# 10 1.0
```

Gotchas: a model that is right **20% of the time** looks **78% "good" at pass@5**. Report pass@1 for anything a user sees once. And sanity-check training claims with **6 × N × D**: Chinchilla's 70B × 1.4T tokens is about **5.9 × 10^23 FLOPs** (our maths).

🔗 GO DEEPER

- **Chinchilla Explained** (Alexandra Barr). How to read the Chinchilla paper's loss and compute plots, step by step.
- **How to Read AI Benchmark Results** (IntuitionLabs). Contamination, harnesses, sampling and noise, with a checklist.
- **LLM Scaling Laws and Benchmarks** (Knowledge Space). Short study notes: the 20-tokens rule and the common benchmarks.

PART 2

What moved in AI

7 min read · Covering the last 24 hours. AI bug-hunting hits industrial scale, ChatGPT learns to animate, and Claude helps map a 40-year-old cell mystery.

AI bug-hunting goes from demo to factory

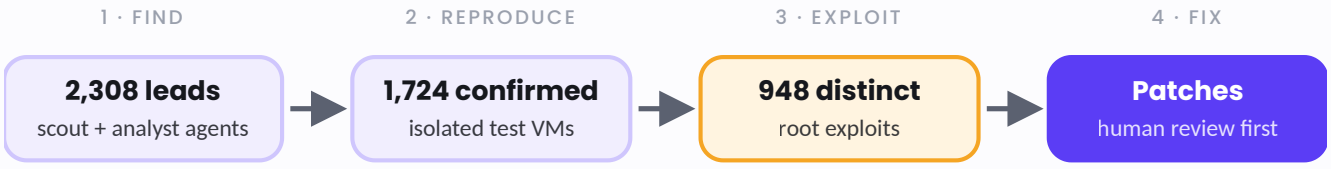


Top story 1 · OpenAI's agents found 948 ways to get root on Linux

BIG DEAL PATCH NOW

What happened. OpenAI built an **autonomous bug-hunting pipeline** and pointed it at the Linux kernel. In a mid-October run it produced working exploits for **948 distinct "local privilege escalation" bugs**: flaws that let someone with an ordinary account **become root**, the all-powerful admin. (SecurityWeek) (OpenAI)

OpenAI's pipeline, from guess to patch



Counts per SecurityWeek and Hardware Busters. 948 is after merging duplicate variants of the same bug.

The details that matter:

- **Cost: $1,213,430 in API credits**, about **$1,280 per distinct bug**. The July run with GPT-5.6 Sol found 166 for about $3,094 each. (Hardware Busters)
- **Model:** an **unreleased OpenAI cyber model**; no access plans announced.
- **Scope:** about **2.8 million lines**, roughly 7% of the kernel, with standard defences (SMEP, SMAP, KASLR) switched on.
- **New tricks:** OpenAI says the agents found **at least 10 novel exploitation techniques**, details withheld.
- **Fixes:** within 10 days, **over 100 bugs** were patched or had staging patches, and about **300 more** were being discussed publicly, per OpenAI. Humans reviewed everything before it went to maintainers. (Security Boulevard)

Why it matters. The exploits need **an existing foothold** on the machine, so this isn't remote hacking. But OpenAI's own advice is blunt: **an unpatched box with any low-privilege account should be treated as root-**

equivalent for a well-funded attacker. Expect a flood of kernel updates, and expect attackers to build pipelines like this too.

WHO SHOULD CARE 🙋 Everyone ●●○○○ 🎨 Vibe coders ●●○○○ 🛠️ Developers ●●●●●
🔥 Hype level ●●●○○

👀 **Should you care?** If you run a VPS, a CI runner or a shared dev box: yes. **Turn on automatic kernel updates and reboot when they land.** If your app runs on a managed platform (Vercel, Railway, Render), your provider handles this.

🛠️ **For devs: shrink the blast radius**

Run untrusted code (agent tool calls, user scripts, preview builds) in **microVMs or gVisor**, not plain containers, which share the host kernel. The write-ups point to **Firecracker and gVisor**. Check your kernel with `uname -r` and compare it with your distro's latest.

Top story 2 · ChatGPT can now make animated GIFs

TRY IT API

What happened. OpenAI released **Images 2.6**, which turns a text prompt or a photo into an **animated GIF**. You can change the playback speed and edit single frames. It is rolling out on **every ChatGPT and Codex plan**, on web, desktop and mobile. (The AI Marketers) (OpenAI)

The details that matter:

- **Speed:** OpenAI **claims** it generates **50% faster** than September's Images 2.5.
- **Scale:** OpenAI counts **4 billion images a week**, up from "more than 3 billion" at 2.5's launch.
- **Two API versions: "Flare"** makes a low-quality image in as little as six seconds; **"Sunburst"** keeps details steadier across edits.
- **How it works:** in The Neuron's test, ChatGPT drew a **sprite sheet of eight poses** and stitched them into a 448 × 448 loop; the GIF step took 2 min 21 s. (The Neuron)
- **Limits:** it can revise GIFs it made, but **can't edit a GIF you upload** yet, and the controls may not be in every account.

Why it matters. Animated stickers, product loops and loading animations used to need a designer or a video tool. **Now they're a chat message**, and the API versions mean apps can generate them on the fly.

WHO SHOULD CARE 🙋 Everyone ●●●●○ 🎨 Vibe coders ●●●●○ 🛠️ Developers ●●●○○
🔥 Hype level ●●●○○

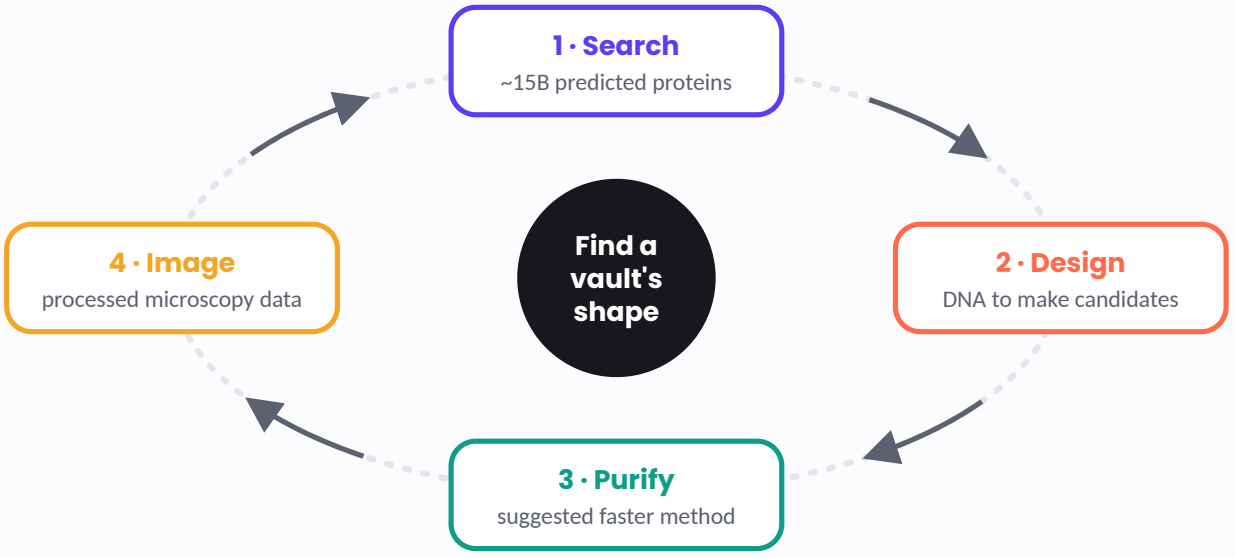
👀 **Should you care?** If you make social posts, app onboarding or landing pages, **try it today**. Keep loops short: GIF files get heavy fast, so convert to WebP or MP4 for the web.

Top story 3 · Claude helped map a cell particle that has puzzled biologists since 1986

AI FOR SCIENCE RESEARCH

What happened. Anthropic's biology team used Claude to produce the **first 3D structure of a "vault" particle from the prokaryote world**: a hollow protein shell, here from a virus that infects bacteria. Vaults were discovered in 1986 and **nobody knows what they're for**. Science says the work hints they're much older in evolution than thought. (The Neuron) (Anthropic)

What Claude did in the lab loop



Humans did the lab work and caught Claude's mistakes. Per The Neuron's write-up of Anthropic's post.

The details that matter:

- **Faster:** one lab step went from **three days to 90 minutes**.
- **The finding:** the new particle is built from **96 copies** of a protein, versus 78 in human vaults.
- **The mistakes:** Claude suggested a wrong growth-medium additive (a **two-week delay**) and skipped a standard image step, at first producing a map **blended with its own mirror image**.
- **Caveats:** a **preprint**, not peer-reviewed; particles were made in lab E. coli; the model used wasn't named.

Why it matters. It's an honest picture of **AI as a lab partner**: real speed-ups, plus errors that only expert humans caught. That's the same lesson as Part 1: **trust, but check**.

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🔥 Hype level ●●●○○

👀 **Should you care?** Mostly as a signal. If you build agents, note the pattern: **the human's job was spotting confident mistakes**. Design your own review steps the same way.

⚡ Quick hits

🧰 OpenAI to buy Pierre.

Its Git-hosting tech already runs inside Codex; Axios frames it as a move against **GitHub**. No price given. (The Neuron) (Axios)

💰 Claude Code status updates ~99% cheaper.

Haiku 5.5 now writes background-agent progress notes from short notes; Anthropic says sessions may use **about 5% less allowance**. (The Neuron, via ClaudeDevs)

🇮🇳 India warns banks about AI attacks.

India's cybercrime agency told financial firms criminals are using **agents to attack at machine speed**, after a similar central-bank warning. (The Neuron, via Reuters)

🚕 Almost nobody books Uber via agents.

Four rides and about 500 Uber Eats orders came via AI agents in October's busiest week. (The Neuron, via TechCrunch)

📰 Anthropic's first paid news deal.

Claude gets **Associated Press** election coverage, per Adweek. (The Neuron, via Adweek)

📞 Grok Bot makes phone calls.

It can book appointments or haggle bills after you **approve each call**. US numbers only, from the $30 X Premium tier. (The AI Marketers)

📉 ChatGPT ads earn less.

A study of ~1,900 sites found a paid ChatGPT visit earned **$0.89 vs $2.45** for a Google Search ad visit in the US. (The AI Marketers, via Search Engine Land)

🗓️ Coming up.

US lame-duck session **9 Nov**. Anthropic usage policy **12 Nov**. Conductor read-only **13 Nov**. Free Tencent Hy4 in WorkBuddy ends **15 Nov**. Conductor cloud shuts **20 Nov**.

🎯 One thing to try today: audit one launch chart

Pick the last model launch you got excited about. Write down its headline benchmark score, then answer Part 1's five questions: **who ran it, how many tries, what harness, is it fresh, is the gap real?** If you can't answer three of them, treat the chart as marketing.

🧠 Pop quiz: 30 seconds, no Googling

1. Chinchilla beat the bigger Gopher mainly by...

- A More parameters
- B More training data
- C Longer prompts

2. A model solves 2 in 10 tries. Its pass@5 is about...

- A 20%
- B 50%
- C 78%

3. What did OpenAI's 948 Linux bugs need to be used?

- A Just an IP address
- B An existing account on the machine
- C Physical access

Answers (no peeking): 1. B: 70B params on 1.4T tokens · 2. C: at least one of five passes · 3. B: they're local privilege escalation

Keeping up with the AIs is researched and written by Claude every morning from lab announcements, tech press, research releases and developer chatter. Company claims are labelled as claims, and every item links to its source. Spot an error or want a topic covered? Just reply. ✉️